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Centre Number

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Student Number

SCEGGS Darlinghurst

2009

**Preliminary Course
Semester 2 Examination**

Mathematics Extension 1

Outcomes Assessed: P1-P8, PE1, PE2, PE3 and PE6

Task Weighting: 40%

General Instructions

- Reading time – 5 minutes
- Time allowed – 1½ hours
- Write using black or blue pen
- Write your Student Number at the top of this page
- This paper has **four** questions
- Attempt all questions on the pad paper provided
- Answer **all** questions and show all necessary working
- **Start each question on a new page**
- Marks will be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used
- **Do not** attach all questions together in one bundle

Total marks – 60

- Attempt Questions 1 – 4

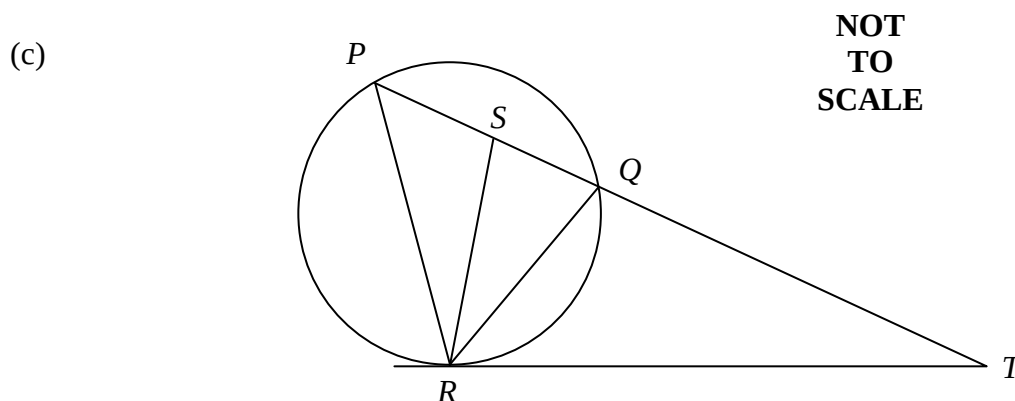
Question	Comm	Reason	Calc	TOTAL
1		/4	/2	/15
2	/3	/4		/16
3		/5		/13
4		/3	/2	/16
TOTAL	/3	/16	/4	/60

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Question 1 (15 marks)

(a) The polynomial $x^3 + 2x^2 + 1$ is divided by $x + 3$. Calculate the remainder. 2

(b) The interval AB , where A is $(4, 5)$ and B is $(19, -5)$ is divided internally in the ratio $2 : 3$ by the point $P(x, y)$. Find the values of x and y . 2



In the diagram ΔPQR is drawn in a circle. The tangent to the circle at R meets PQ produced to T . S is a point on PQ such that RS bisects $\angle QRP$.

Copy the diagram.

(i) Explain why $\angle TRQ = \angle RPT$. 1

(ii) Show that $TR = TS$, giving clear reasons. 3

(d) A monic polynomial $P(x)$ of degree 4 has one root equal to 2 and a double root equal to -1 . 2

If the polynomial passes through the point $(0, 6)$ find the equation of $P(x)$.

Question 1 continues on the next page

Question 1 (continued)

- (e) Let $f(x) = 3x^2 - 2x$. **2**

Use the definition $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$

to find the derivative by first principles of $f(x)$ at the point $x = a$.

- (f) By making an appropriate substitution, or otherwise, solve for x . **3**

$$2\left(x + \frac{1}{x}\right)^2 + \left(x + \frac{1}{x}\right) - 15 = 0$$

- Start a new page

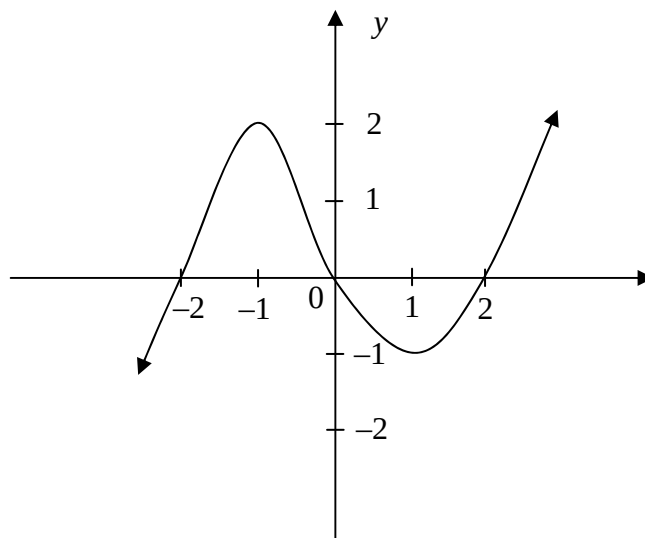
Marks

Question 2 (16 marks)

- (a) Solve for θ , $0^\circ \leq \theta \leq 360^\circ$ **3**

$$\sin 2\theta = \sin \theta$$

- (b) The curve $y = f(x)$ is shown below.



On the separate answer page attached, draw the graph of:

(i) $y = |f(x)|$ **1**

(ii) $y = f(x + 2)$ **1**

(iii) $y = f(x) - 1$ **1**

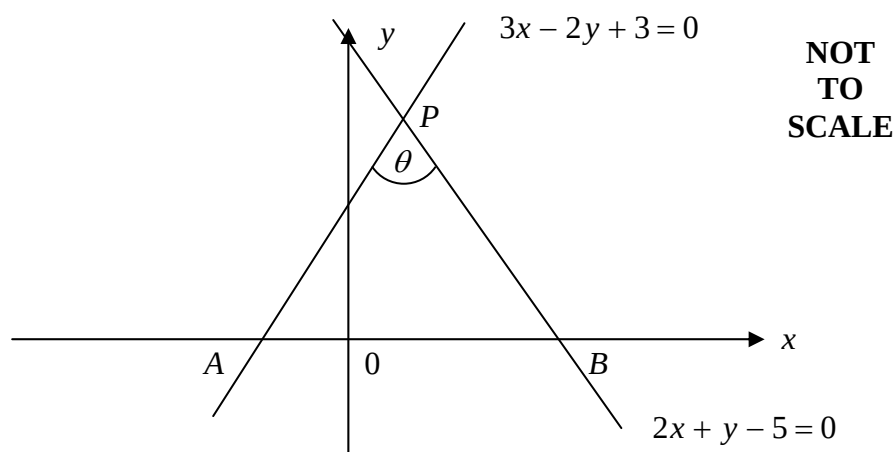
Question 2 continues on the next page

Question 2 (continued)

- (c) The polynomial $P(x) = 2x^3 + 3x^2 + kx - 2$ has roots α , β and γ .
- (i) Find the value of $\alpha + \beta + \gamma$. **1**
- (ii) Find the value of $\alpha\beta\gamma$. **1**
- (iii) If one root is the reciprocal of the other, find the third root and hence find the value of k . **2**
- (d) (i) Express $\sin x + \cos x$ in the form $R \cos(x - \alpha)$ where $R > 0$ and $0^\circ < \alpha < 90^\circ$. **2**
- (ii) Hence solve for $0^\circ \leq x \leq 360^\circ$ **2**
- $$\sin x + \cos x = 1$$
- (iii) Find all possible solutions of $\sin x + \cos x = 1$. **2**

Question 3 (13 marks)

(a)



The diagram shows the lines $3x - 2y + 3 = 0$ and $2x + y - 5 = 0$ intersecting at point P . The lines cut the x and y axes at A and B respectively.

(i) Find the co-ordinates of P . 2

(ii) Find the size of the acute angle, θ , between the lines $3x - 2y + 3 = 0$ and $2x + y - 5 = 0$. Answer correct to the nearest minute. 3

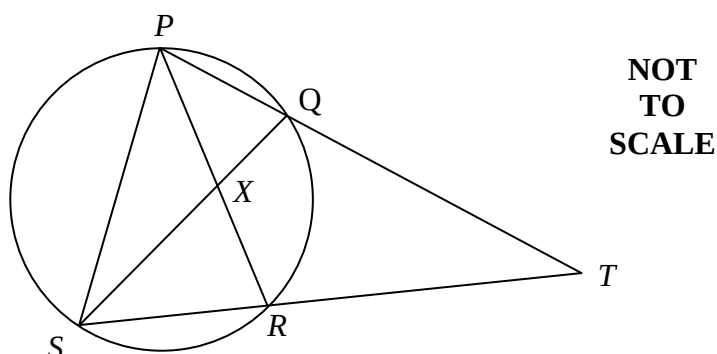
(b) (i) By using the substitution $t = \tan \frac{x}{2}$, show that 2
 $\sin x - 3\cos x - 2$
 can be written as $\frac{t^2 + 2t - 5}{1 + t^2}$.

(ii) Hence, find correct to the nearest minute 3
 the values of x , $0^\circ \leq x \leq 360^\circ$ for which
 $\sin x - 3\cos x - 2 = 0$.

Question 3 continues on the next page

Question 3 (continued)

(c)



The points P, Q, R, S are placed on a circle of radius r such that PR and QS meet at X .

The lines PQ and SR are produced to meet at T , and $QXRT$ is a cyclic quadrilateral.

Copy or trace this diagram onto your answer page.

(i) Find the size of $\angle SQT$, giving reasons for your answer. 2

(ii) Find an expression for the length of PS in terms of r . 1

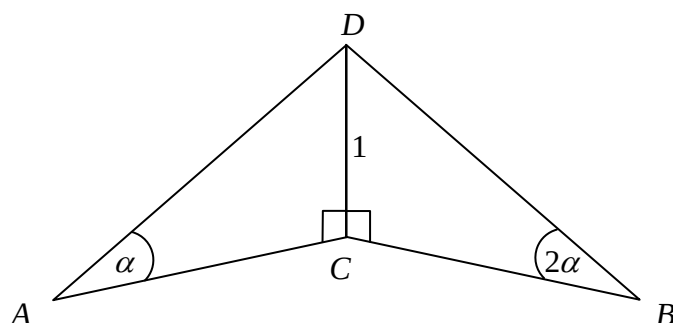
Question 4 (16 marks)

- (a) Solve for x .

3

$$\frac{3}{2x-1} \geq x$$

- (b)



In the diagram CD is a vertical flagpole of height 1 metre which stands on horizontal ground.

The angles of elevation of the top D of the flagpole from points A and B on the ground are α and 2α respectively.

- (i) Show that $BC = \frac{1 - \tan^2 \alpha}{2 \tan \alpha}$.

1

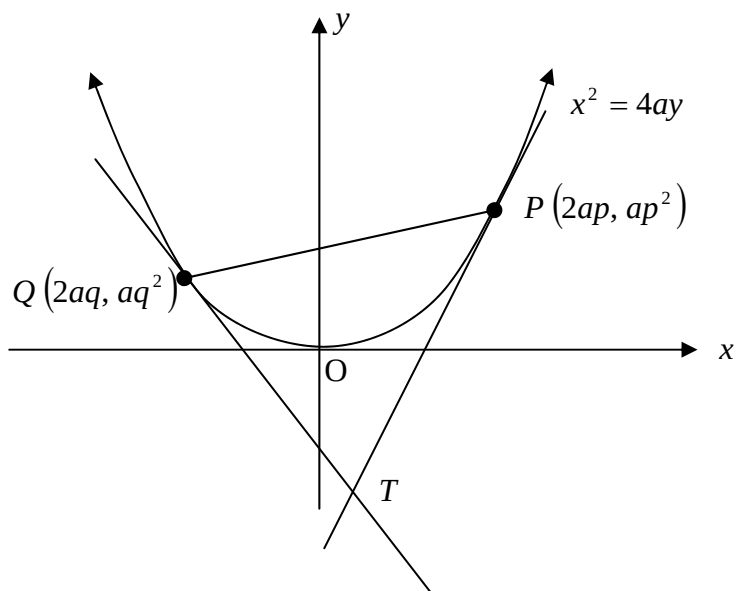
- (ii) Show that $AC - BC = BD$.

3

Question 4 continues on the next page

Question 4 (continued)

- (c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.
The tangents at P and Q intersect at T .



- (i) Show that the tangent at P is given by $y = px - ap^2$. 2
- (ii) Show that the tangents intersect at the point $T(a(p + q), apq)$. 2
- (iii) Show that the chord PQ has equation $y = \frac{1}{2}(p + q)x - apq$. 2
- (iv) The line PQ is a tangent to the parabola $x^2 = 2ay$. 2
- Show that $(p + q)^2 = 8pq$.
- (v) Show that the Cartesian equation of the locus of T is a parabola. 1

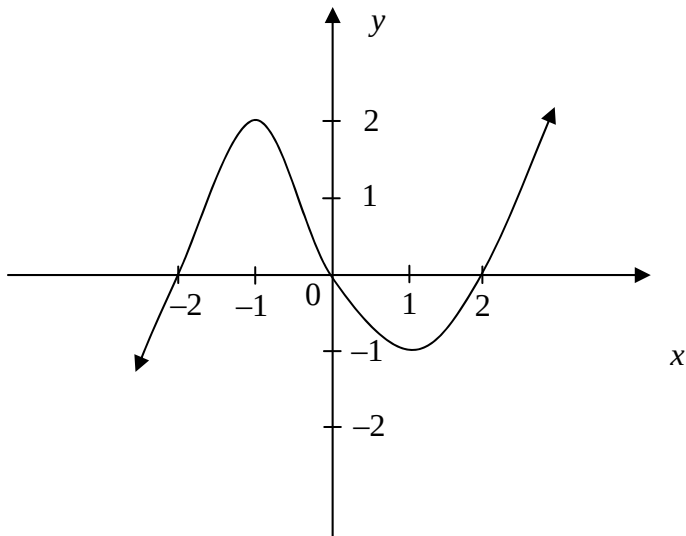
End of paper

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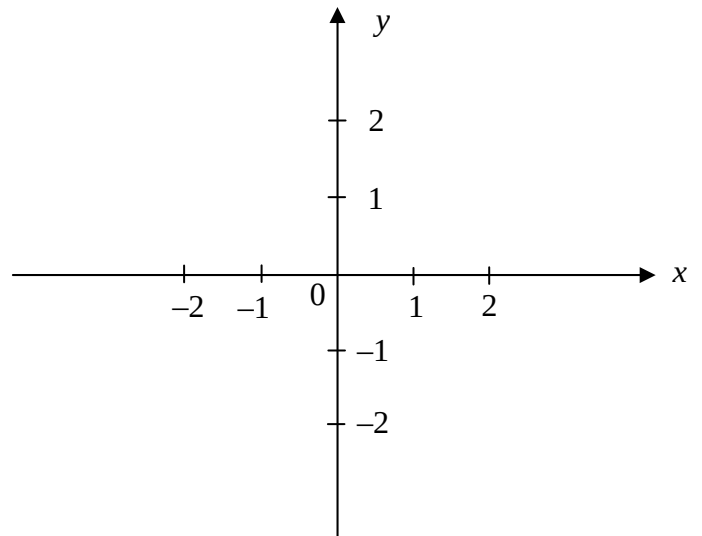
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Use this page to answer Question 2 (b)

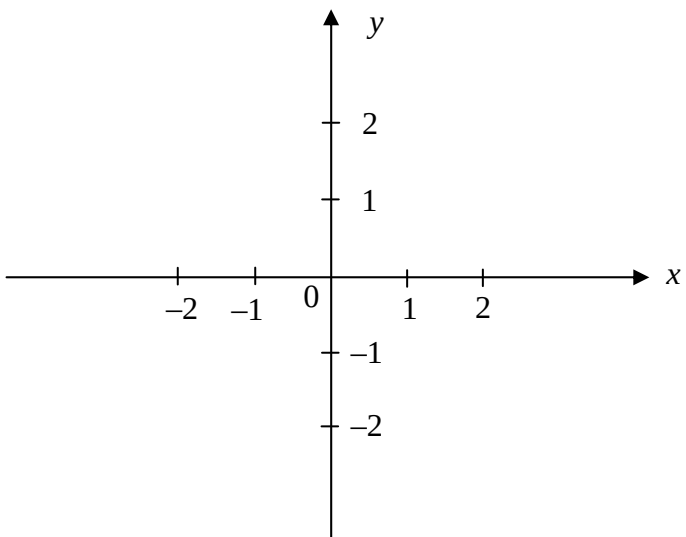
$$y = f(x)$$



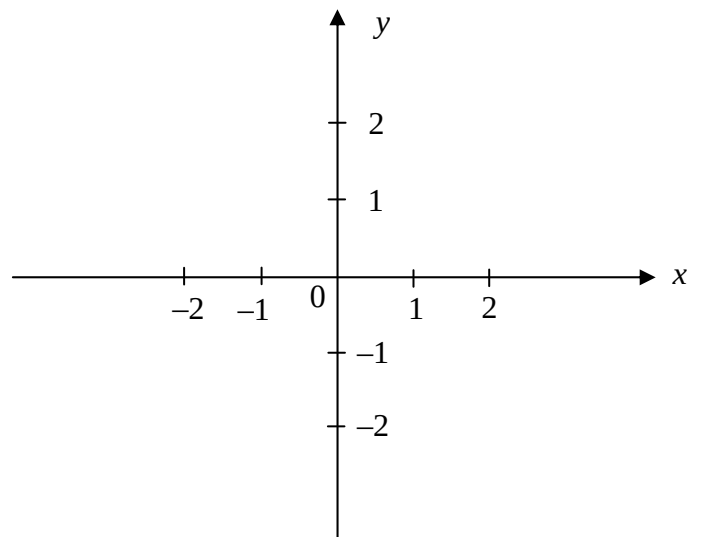
(i) $y = |f(x)|$



(ii) $y = f(x + 2)$



(iii) $y = f(x) - 1$



Year 11 Extension 1 Mathematics

Preliminary Semester 2 Exam 2009

Comments

Question 1

a) $P(x) = x^3 + 2x^2 + 1$

Remainder

$$\begin{aligned} P(-3) &= (-3)^3 + 2(-3)^2 + 1 \quad \checkmark \\ &= -27 + 18 + 1 \\ &= -8 \quad \checkmark \end{aligned}$$

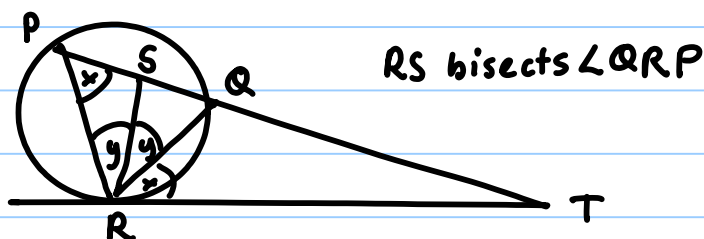
b) $A(4, 5) \quad B(19, -5)$

$2:3$ Internal division

$$P = \left(\frac{3 \times 4 + 2 \times 19}{2 + 3}, \frac{3 \times 5 + 2 \times -5}{2 + 3} \right)$$

$$\begin{aligned} &= \left(\frac{50}{5}, \frac{5}{5} \right) \quad \checkmark \checkmark \\ &= (10, 1) \end{aligned}$$

c)



i) $\angle TRQ = \angle TPR$

(Angle between a tangent and a chord is equal to the angle in the alternate segment.) $\checkmark$

ii) Let $\angle TRQ = \angle TPR = x$

Let $\angle QRS = \angle SRP = y$ (equal angles since RS bisects $\angle QRP$)

$\angle TSR = \angle SPR + \angle SRP$ (exterior angle of a triangle is equal to sum of opposite interior angles) $\checkmark$
 $= x + y$

$\angle SRT = x + y$

$\therefore TR = TS$ (sides opposite equal angles are equal.) $\checkmark \checkmark$

(Reas 4)

Question 1 continued.

d)

$$P(x) = (x-2)(x+1)^2(x+k)$$

↑
monic
coefficient of x^4 is one

↑
must put extra
factor because
of degree 4

passes through $(0,6)$

$$6 = 2 \times (1)^2 \times k$$

$$-2k = 6$$

$$k = -3$$

e)

$$f(x) = 3x^2 - 2x$$

$$f(a) = 3a^2 - 2a$$

$$f(a+h) = 3(a+h)^2 - 2(a+h)$$

$$= 3(a^2 + 2ah + h^2) - 2a - 2h$$

$$= 3a^2 + 6ah + 3h^2 - 2a - 2h$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3a^2 + 6ah + 3h^2 - 2a - 2h - (3a^2 - 2a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6ah + 3h^2 - 2h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(6a + 3h - 2)}{h}$$

$$= \lim_{h \rightarrow 0} (6a + 3h - 2)$$

$$= 6a - 2$$

Question 1 (continued)

f) $2\left(x + \frac{1}{x}\right)^2 + \left(x + \frac{1}{x}\right) - 15 = 0$

Let $m = x + \frac{1}{x}$

$$2m^2 + m - 15 = 0$$

$$(2m - 5)(m + 3) = 0$$

$$m = \frac{5}{2}$$

$$x + \frac{1}{x} = \frac{5}{2}$$

$$x^2 + 1 = \frac{5}{2}x$$

$$2x^2 + 2 = 5x$$

$$2x^2 - 5x + 2 = 0$$

$$(2x - 1)(x - 2) = 0$$

$$x = \frac{1}{2}, 2$$

$$m = -3$$

$$x + \frac{1}{x} = -3$$

$$x^2 + 1 = -3x$$

$$x^2 + 3x + 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-3 \pm \sqrt{9 - 4 \cdot 1 \cdot 1}}{2}$$

$$= \frac{-3 \pm \sqrt{5}}{2}$$

Question 2

a) $\sin 2x = \sin x$

$$2\sin x \cos x - \sin x = 0$$

$$\sin x (2\cos x - 1) = 0$$

$$\sin x = 0$$

$$x = 0^\circ, 180^\circ, 360^\circ$$

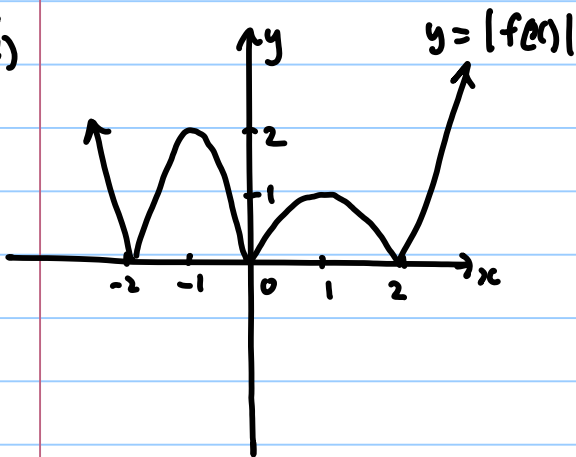
$$\cos x = \frac{1}{2}$$

quad 1 & 4

$$x = 60^\circ, 300^\circ$$

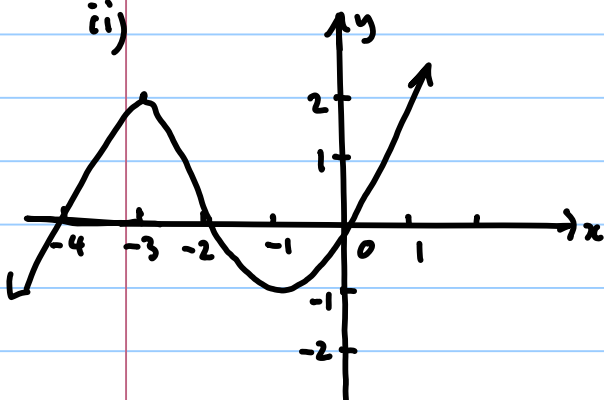
Many divided this line by $\sin x$, doing this eliminates one of the resulting equations. You must factorise then solve each equation. Also remember that 360° is in the domain

b) i)



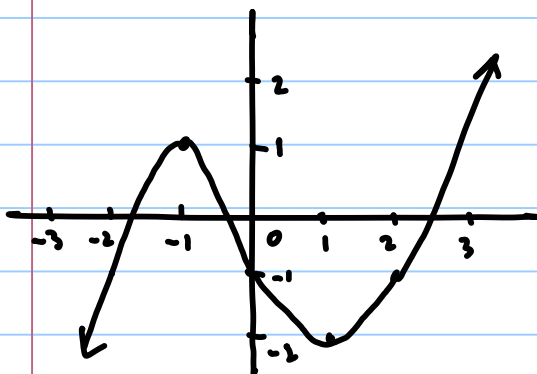
These were well done

ii)



$$y = f(x+2)$$

Shift left 2 units.



$$y = f(x) - 1$$

Shift down 1 unit

If your curve is shifted down then the intercepts will also change

Question 2 (continued)

c) $P(x) = 2x^3 + 3x^2 + kx - 2$

i) $\alpha + \beta + \gamma = -\frac{b}{a}$
 $= -\frac{3}{2}$

ii) $\alpha\beta\gamma = -\frac{d}{a}$
 $= \frac{2}{2}$
 $= 1$

iii) One root is reciprocal $\Rightarrow$ Roots are $\alpha, \frac{1}{\alpha}, \gamma$

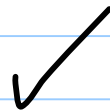
from part ii)

$$\alpha\beta\gamma = 1$$

$$\alpha \cdot \frac{1}{\alpha} \gamma = 1$$

$$\gamma = 1$$

R
(2)



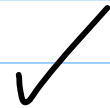
Since this is a root of the polynomial $P(1) = 0$

$$2x(1)^3 + 3x(1)^2 + kx - 2 = 0$$

$$2 + 3 + k - 2 = 0$$

$$k + 3 = 0$$

$$k = -3$$



Parts (i) and (ii) were well done.

Read the question carefully it says one root is the reciprocal of the other, not the negative reciprocal

Question 2 (continued)

d) i)

$$\begin{aligned}\sin x + \cos x &= R \cos(x - \alpha) \\ &= R(\cos x \cos \alpha + \sin x \sin \alpha) \\ &= R \cos x \cos \alpha + R \sin x \sin \alpha\end{aligned}$$

Part (i) well done

Match The coefficients

$$R \cos \alpha = 1 \quad \textcircled{1}$$

$$R \sin \alpha = 1 \quad \textcircled{2}$$

Find R

$$R^2 = 1^2 + 1^2$$

$$\begin{aligned}R &= \sqrt{1+1} \\ &= \sqrt{2}\end{aligned}$$

Find α

$$\textcircled{2} / \textcircled{1}$$

$$\tan \alpha = 1$$

$$\alpha = 45^\circ$$

$$\therefore \sin x + \cos x = \sqrt{2} \cos(x - 45^\circ)$$

ii)

$$\sin x + \cos x = 1$$

$$\sqrt{2} \cos(x - 45^\circ) = 1$$

$$\cos(x - 45^\circ) = \frac{1}{\sqrt{2}}$$

quad 1 & 4

Check domain

$$0^\circ \leq x \leq 360^\circ$$

$$-45^\circ \leq x - 45^\circ \leq 315^\circ$$

$$\therefore x - 45^\circ = -45^\circ, 45^\circ, 360^\circ - 45^\circ$$

↑
quad 4

↑
quad 1

↑
quad 4

$$x = 0^\circ, 90^\circ, 360^\circ$$

✓✓

Remember to check the domain, often a solution is left out if this is not done

iii) all solutions → general solution

$$\cos(x - 45^\circ) = \frac{1}{\sqrt{2}}$$

R
②

$$x - 45^\circ = 360^\circ n \pm \cos^{-1} \frac{1}{\sqrt{2}}$$

$$x = 360^\circ n \pm 45^\circ + 45^\circ$$

$$\therefore x = 360^\circ n \text{ or } x = 360^\circ n + 90^\circ$$

where $n \in \mathbb{Z}$

✓

Not well done, all possible solutions means general solution
Two methods

- learn the rule

- look for a pattern

Another Method

$$\cos(x - 45^\circ) = \frac{1}{\sqrt{2}}$$

quadrants 1 or 4

$$\text{in Q1: } x - 45^\circ = 45^\circ, 360^\circ + 45^\circ, 2 \times 360^\circ + 45^\circ, \dots$$

$$x = 90^\circ, 360^\circ + 90^\circ, 2 \times 360^\circ + 90^\circ$$

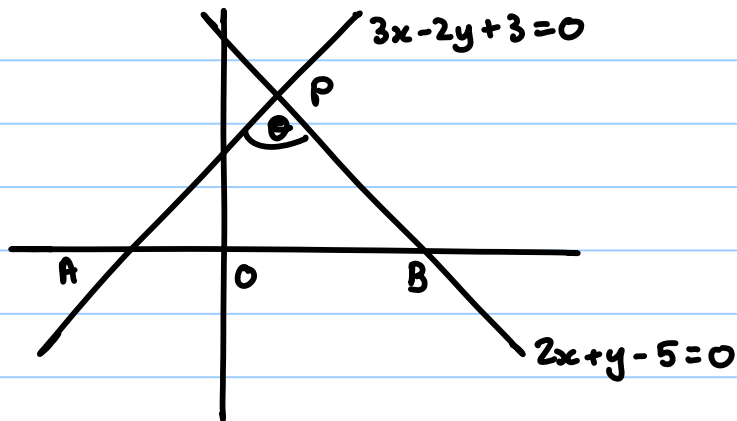
$$\text{Q4: } x - 45^\circ = 360^\circ - 45^\circ, 2 \times 360^\circ - 45^\circ, \dots$$

$$x = 360^\circ, 2 \times 360^\circ, \dots$$

$$\therefore x = 360^\circ n + 90^\circ \quad \text{or} \quad x = 360^\circ n$$

Question 3

a)



$$\text{i) } 3x - 2y + 3 = 0 \quad (1)$$

$$2x + y - 5 = 0 \quad (2)$$

$$(2) \times 2 \quad 4x + 2y - 10 = 0 \quad (3)$$

$$3x - 2y + 3 = 0 \quad (1)$$

$$\text{Add } (1) \text{ and } (3) \quad 7x - 7 = 0$$

$$7x = 7$$

$$x = 1$$

$$\text{Subst. into } (1) \quad 3 - 2y + 3 = 0$$

$$2y = 6$$

$$y = 3 \quad \therefore P(1, 3)$$

$$\text{ii) } 3x - 2y + 3 = 0$$

$$m_1 = -\frac{3}{-2} \\ = \frac{3}{2}$$

$$2x + y - 5 = 0$$

$$m_2 = -\frac{2}{1} \\ = -2$$

$$\text{Acute angle } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{3}{2} - (-2)}{1 + \frac{3}{2} \times (-2)} \right|$$

$$= \left| \frac{3\frac{1}{2}}{-2} \right|$$

$$= \left| -\frac{7}{4} \right|$$

$$\tan \theta = \frac{7}{4}$$

$$\theta = 60^\circ 15'$$

$$ax + by + c = 0$$

To find gradient without making y the subject

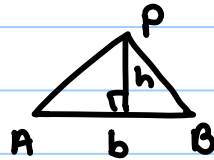
$$m = -\frac{a}{b}$$

Question 3 (continued)

Reas 2

Hence Use $\text{Area} = \frac{1}{2} ab \sin C$

or otherwise



use $A = \frac{1}{2} bh$

You need to find points A, B and some lengths.

Point A put $y=0$ into $3x-2y+3=0$

$$3x + 3 = 0$$

$$3x = -3$$

$$x = -1$$

$A(-1, 0)$

Point B put $y=0$ into $2x+y-5=0$

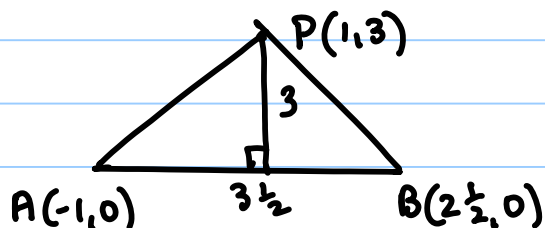
$$2x - 5 = 0$$

$$2x = 5$$

$$x = 2\frac{1}{2}$$

$B(2\frac{1}{2}, 0)$

or otherwise is less work!



By counting horizontally and vertically

$$\text{base} = 3\frac{1}{2}$$

perpendicular height = 3 units

$$\begin{aligned} \therefore \text{Area } \triangle APB &= \frac{1}{2} \times 3\frac{1}{2} \times 3 \\ &= 5.25 \text{ units}^2 \end{aligned}$$

Question 3 (continued)

b) i) $t = \tan \frac{x}{2}$

$$\sin x = \frac{2t}{1+t^2} \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\sin x - 3 \cos x - 2 = \frac{2t}{1+t^2} - 3 \left(\frac{1-t^2}{1+t^2} \right) - 2$$

$$= \frac{2t - 3 + 3t^2 - 2(1+t^2)}{1+t^2}$$

$$= \frac{2t - 3 + 3t^2 - 2 - 2t^2}{1+t^2}$$

$$= \frac{t^2 + 2t - 5}{1+t^2}$$

ii) $\sin x - 3 \cos x - 2 = 0$
 $\frac{t^2 + 2t - 5}{1+t^2} = 0$

$$t^2 + 2t - 5 = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{4 - 4 \times 1 \times -5}}{2}$$

$$= \frac{-2 \pm \sqrt{24}}{2}$$

$$t = \frac{-2 + \sqrt{24}}{2}$$

$$\tan \frac{x}{2} = 1.449...$$

Quad 1 & 3
 (but Quad 3 angle will be too big when doubled)

$$\frac{x}{2} = 55^\circ 24'$$

$$x = 110^\circ 48'$$

$$t = \frac{-2 - \sqrt{24}}{2}$$

$$\tan \frac{x}{2} = -3.449...$$

(Quad 2 and 4
 but Quad 4 angle will be too big)

$$\frac{x}{2} = (180^\circ - 73^\circ 50')$$

$$\frac{x}{2} = 106^\circ 10'$$

$$x = 212^\circ 20'$$

Check domain

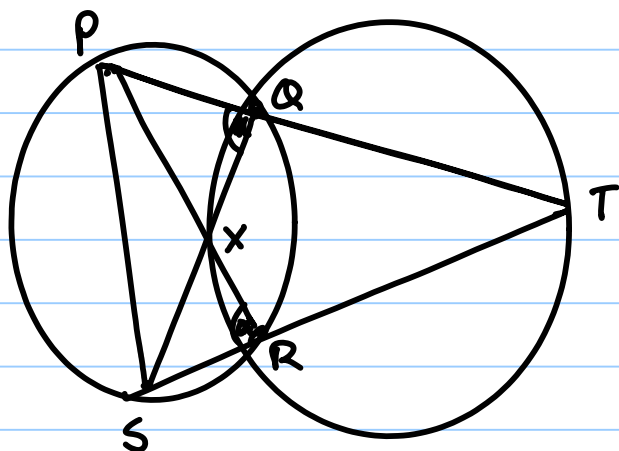
$$0^\circ \leq x \leq 360^\circ$$

$$0^\circ \leq \frac{x}{2} \leq 180^\circ$$

So look for answers in Quadrants 1 & 2 only.

Question 3 (continued)

Reas 3



- i) Let $\angle PQS = \angle PRS = x$
(angles in the same segment are equal)

$$\angle XQT = 180^\circ - \angle PQS = 180^\circ - x$$

$$\angle TRP = 180^\circ - \angle PRS = 180^\circ - x$$

(Angle sum of a straight line is 180°)

$$\angle XQT = \angle TRP$$

Since QXRT is a cyclic quadrilateral

$$\angle XQT + \angle TRX = 180^\circ$$

$$\therefore 2\angle XQT = 180^\circ$$

$$\angle XQT = 90^\circ$$

$$\therefore \angle SQT = 90^\circ$$

- ii) Since $\angle SQT = 90^\circ$

$$\angle SQP = 90^\circ$$

(Angle sum straight line is 180°)

$\therefore$ PS is a diameter of the circle

(Angle subtended by the diameter is 90° at the circumference)

Since r is the radius of the circle

$$PS = 2r$$

Question 4

a) $\frac{3}{2x-1} \geq x$

Not well done

undefined when $x = \frac{1}{2}$



multiply both sides by $(2x-1)^2$

$$\frac{3}{2x-1} \times (2x-1)^2 \geq x(2x-1)^2$$

$$3(2x-1) \geq x(2x-1)^2$$

$$3(2x-1) - x(2x-1)^2 \geq 0$$

don't expand.

factorise it instead!

$$(2x-1)[3 - x(2x-1)] \geq 0$$

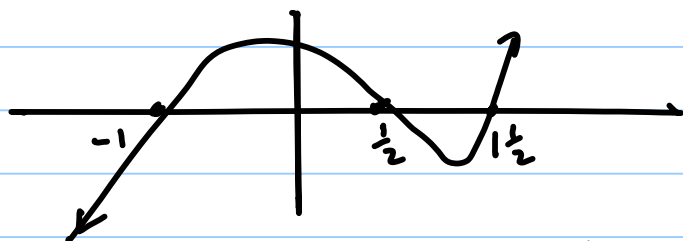
$$(2x-1)(3 - 2x^2 + x) \geq 0$$

↑ factorise out -1 and then reverse signs

$$(2x-1) \cdot -1(2x^2 - x - 3) \geq 0$$

$$(2x-1)(2x^2 - x - 3) \leq 0$$

$$(2x-1)(2x-3)(x+1) \leq 0$$



$$x \leq -1, -\frac{1}{2} \leq x \leq \frac{3}{2}$$

but $x \neq \frac{1}{2} \therefore$ The solution is

$$x \leq -1, -\frac{1}{2} < x \leq \frac{3}{2}$$



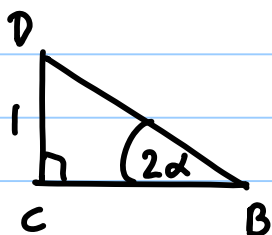
most students chose to expand at this point although this can be done it takes a lot more time and effort

Please factorise, it is much more efficient and you will be less likely to make a mistake

make sure you answer the question and incorporate any undefined values

Question 4 (continued)

b)



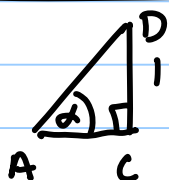
$$\tan 2\alpha = \frac{1}{BC}$$

$$\begin{aligned} BC &= \frac{1}{\tan 2\alpha} \\ &= \frac{1}{\frac{2\tan\alpha}{1-\tan^2\alpha}} \\ &= \frac{1-\tan^2\alpha}{2\tan\alpha} \end{aligned}$$

✓

Part (i) was well done

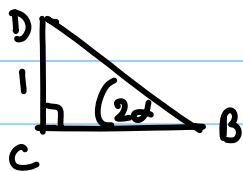
ii)



$$\tan\alpha = \frac{1}{AC}$$

$$AC = \frac{1}{\tan\alpha}$$

R
②



$$\sin 2\alpha = \frac{1}{BD}$$

$$BD = \frac{1}{\sin 2\alpha}$$

This was a tricky question. You needed to establish an expression for AC and BD in terms of α then set the question up as a trig identity proof.

Problems arose when students wrote an expression for BD using pythagoras - although this can be done it makes the question very difficult.

✓

Question 4 (continued)

Prove that $AC - BC = BD$

$$AC - BC = \frac{1}{\tan \alpha} - \left(\frac{1 - \tan^2 \alpha}{2 \tan \alpha} \right)$$

$$= \frac{2 - 1 + \tan^2 \alpha}{2 \tan \alpha}$$

$$= \frac{1 + \tan^2 \alpha}{2 \tan \alpha}$$

$$= \frac{\sec^2 \alpha}{2 \tan \alpha}$$

$$= \frac{1}{\cos^2 \alpha} \div \frac{2 \sin \alpha}{\cos \alpha}$$

$$= \frac{1}{\cos^2 \alpha} \times \frac{\cos \alpha}{2 \sin \alpha}$$

$$= \frac{1}{2 \sin \alpha \cos \alpha}$$

$$= \frac{1}{\sin 2\alpha}$$

$$= BD$$

✓ you needed to use
 $1 + \tan^2 x = \sec^2 x$

✓

Question 4 (continued)

i) $x^2 = 4ay$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

At $P(2ap, ap^2)$

$$m_T = \frac{2ap}{2a}$$

$$= p$$

Eqn tangent

$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$



Calc
(2)

Parts (i) to (iii) were quite good

* Important note: if a question says show then you must show working eg. don't just write $m_T = p$ you must differentiate and substitute.

ii) Similarly tangent at Q

$$y = qx - aq^2$$

$$y = px - ap^2$$

Solve simultaneously

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p - q) = a(p^2 - q^2)$$

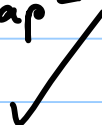
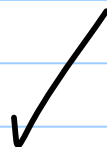
$$x = a \frac{(p - q)(p + q)}{p - q}$$

$$x = a(p + q)$$

subst. into (i)

$$\begin{aligned} y &= ap(p + q) - ap^2 \\ &= ap^2 + apq - ap^2 \\ &= apq \end{aligned}$$

$$T(a(p + q), apq)$$

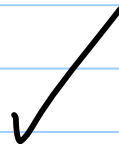


Question 4 (continued)

iii) $P(2ap, ap^2)$ $Q(2aq, aq^2)$

Gradient PQ

$$\begin{aligned} m_{PQ} &= \frac{aq^2 - ap^2}{2aq - 2ap} \\ &= \frac{a(q^2 - p^2)}{2a(q - p)} \\ &= \frac{a(q - p)(q + p)}{2a(q - p)} \\ &= \frac{p + q}{2} \end{aligned}$$



Equation of chord PQ

$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = \left(\frac{p+q}{2}\right)(x - 2ap)$$

$$y - ap^2 = \frac{1}{2}(p+q)x - \frac{1}{2}(p+q) \times 2ap$$

$$y - ap^2 = \frac{1}{2}(p+q)x - ap^2 - apq$$

$$y = \frac{1}{2}(p+q)x - apq$$



Alternate Method

Chord of contact: PQ is a chord of contact from the external point T $(a(ptq), apq)$

$$xx_0 = 2a(y + y_0)$$

$$xa(ptq) = 2a(y + apq)$$

$$x(ptq) = 2y + 2apq$$

$$2y = x(ptq) + 2apq$$

$$y = \frac{1}{2}(ptq)x + apq$$

Question 4 (continued)

(iv) $x^2 = 2ay$ (1) R
 $y = \frac{1}{2}(p+q)x - apq$ (2) 2

Solve simultaneously

$$x^2 = 2a \left(\frac{1}{2}(p+q)x - apq \right)$$

$$x^2 = a(p+q)x - 2a^2pq$$

$$x^2 - a(p+q)x + 2a^2pq = 0$$

This is a quadratic equation.

Since the line PQ is a tangent to the parabola there is only one solution

$\therefore$ Using the discriminant $\Delta = 0$
 $\Delta = b^2 - 4ac$

$$= (-a(p+q))^2 - 4 \cdot 1 \cdot 2a^2pq$$

$$= a^2(p+q)^2 - 8a^2pq$$

$$= a^2 [(p+q)^2 - 8pq]$$

Since $\Delta = 0$

$$a^2 [(p+q)^2 - 8pq] = 0$$

$$(p+q)^2 - 8pq = 0$$

$$(p+q)^2 = 8pq$$

This was a tricky question:

we have the line PQ and the parabola $x^2 = 2ay$. Start by solving these simultaneously

* This is essential to solve this question

Question 4 (continued)

v) $T(a(p+q), apq)$

$$x = a(p+q) \Rightarrow p+q = \frac{x}{a}$$

$$y = apq \Rightarrow pq = \frac{y}{a}$$

You need to eliminate $p+q$.

Use your x and y coordinates to get an expression for $p+q$ and for pq

Using part iv)

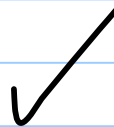
$$(p+q)^2 = 8pq$$

$$\left(\frac{x}{a}\right)^2 = 8\frac{y}{a}$$

$$\frac{x^2}{a^2} = \frac{8y}{a}$$

$$x^2 = \frac{8y}{a} \times a^2$$

$$x^2 = 8ay$$



Substitute these expressions into the equation from part (v)

The locus of T is a parabola.